2314 Honors Project

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Introduction

My goal is to use statistical analysis and visualizations to show a relationship between people aged 18-20, and the use of alcohol and marijuana. I am doing this to see if over this three-year period one of these illegal drugs until 21 has a higher usage rate from year to year over the three-year period. I expect for the usage to be similar over those three years and for the average use of both to be similar as well.

```
#First, I installed and loaded all the packages needed to complete the project.
```

```
pacman::p_load(pacman, corrplot, tidyverse, rio, psych, tsibble, lubridate, ggplot2, gridExtra)
```

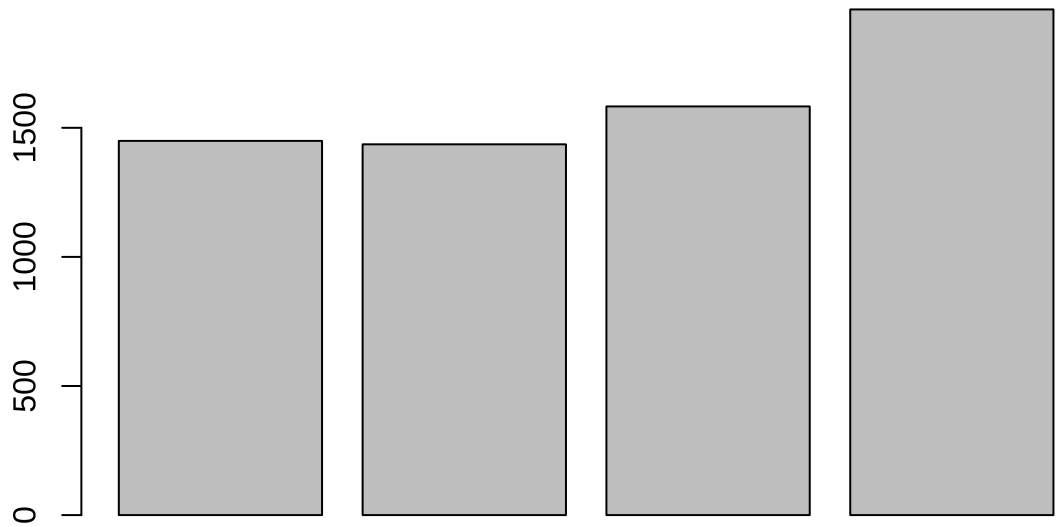
```
#Next, I imported the csv data set as a tibble, and filtered the data to look at only the
#necessary information.
```

```
df <- import("~/INEG 2314/drug-use-by-age.csv") %>%
  as_tibble() %>%
    filter(age >= 18) %>%
  filter(age <= 21) %>%
    mutate(
    `alcohol-use` = (`alcohol-use`/100) * `n`,
    `marijuana-use` = (`marijuana-use`/100) * `n`) %>%
  select(c("age","n","alcohol-use", "alcohol-frequency", "marijuana-use", "marijuana-frequency")) %>%
  print()
```

```
## # A tibble: 4 x 6
##   age       n `alcohol-use` `alcohol-frequency` `marijuana-use`
##   <chr> <int>         <dbl>               <dbl>           <dbl>
## 1 18     2469         1449.                  24            832.
## 2 19     2223         1436.                  36            742.
## 3 20     2271         1583.                  48            772.
## 4 21     2354         1959.                  52            777.
## # i 1 more variable: `marijuana-frequency` <dbl>
```

```
#I then used different ways to explore the dataset and see if there was any interesting
#connections between the data.
```

```
barplot(df$`alcohol-use`)
```




```r
df %>% summary()
```

```
##      age                  n         alcohol-use   alcohol-frequency
##  Length:4           Min.   :2223   Min.   :1436   Min.   :24
##  Class :character   1st Qu.:2259   1st Qu.:1446   1st Qu.:33
##  Mode  :character   Median :2312   Median :1516   Median :42
##                     Mean   :2329   Mean   :1607   Mean   :40
##                     3rd Qu.:2383   3rd Qu.:1677   3rd Qu.:49
##                     Max.   :2469   Max.   :1959   Max.   :52
##  marijuana-use   marijuana-frequency
##  Min.   :742.5   Min.   :52
##  1st Qu.:764.7   1st Qu.:52
##  Median :774.5   Median :56
##  Mean   :780.9   Mean   :56
##  3rd Qu.:790.6   3rd Qu.:60
##  Max.   :832.1   Max.   :60
```

```r
df %>% describe()
```

```
##                     vars n    mean     sd  median trimmed    mad     min
## age*                   1 4    2.50   1.29    2.50    2.50   1.48    1.00
## n                      2 4 2329.25 107.74 2312.50 2329.25  97.11 2223.00
## alcohol-use            3 4 1606.69 243.75 1516.10 1606.69 108.84 1436.06
## alcohol-frequency      4 4   40.00  12.65   42.00   40.00  11.86   24.00
## marijuana-use          5 4  780.87  37.35  774.48  780.87  25.45  742.48
```

```
## marijuana-frequency    6 4   56.00   4.62   56.00   56.00   5.93   52.00
##                         max  range  skew kurtosis     se
## age*                   4.00   3.00  0.00    -2.08   0.65
## n                   2469.00 246.00  0.27    -2.03  53.87
## alcohol-use         1958.53 522.47  0.60    -1.81 121.88
## alcohol-frequency     52.00  28.00 -0.24    -2.12   6.32
## marijuana-use        832.05  89.57  0.37    -1.84  18.68
## marijuana-frequency   60.00   8.00  0.00    -2.44   2.31
```

```r
#From the exploratory data I was able to come up with differences in the usage of
#alcohol and marijuana from year-to-year for 18 to 21 year olds.
```

```r
alc_18_21 <- df %>%
    select(c("alcohol-use", "n"))
alc_18_21
```

```
## # A tibble: 4 x 2
##   `alcohol-use`     n
##           <dbl> <int>
## 1         1449.  2469
## 2         1436.  2223
## 3         1583.  2271
## 4         1959.  2354
```

```r
alc_test_1819 <- prop.test(x = c(1449.303, 1436.058), n = c(2469, 2223))
alc_test_1920 <- prop.test(x = c(1436.058, 1582.887), n = c(2223, 2271))
alc_test_2021 <- prop.test(x = c(1582.887, 1958.528), n = c(2271, 2354))
# Printing the results

alc_test_1819
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(1449.303, 1436.058) out of c(2469, 2223)
## X-squared = 16.949, df = 1, p-value = 3.84e-05
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.08721899 -0.03078101
## sample estimates:
## prop 1 prop 2
##  0.587  0.646
```

```r
alc_test_1920
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(1436.058, 1582.887) out of c(2223, 2271)
## X-squared = 13.021, df = 1, p-value = 0.000308
## alternative hypothesis: two.sided
## 95 percent confidence interval:
```

```
##  -0.07887524 -0.02312476
## sample estimates:
## prop 1 prop 2
##  0.646  0.697
```

```r
alc_test_2021
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(1582.887, 1958.528) out of c(2271, 2354)
## X-squared = 116.67, df = 1, p-value < 2.2e-16
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.1596262 -0.1103738
## sample estimates:
## prop 1 prop 2
##  0.697  0.832
```

```r
#From the data we can see that the p-value is always below the alpha 0.05, so we can
#conclude that for every year, from 18 to 21, that the average usage of alcohol will
#increase. This is is because all of the numbers in our interval are negative meaning
#that the average from the year above is larger than that below. Since there is a
#statistical difference each year the data is increasing at a fast pace.
```

```r
mar_18_21 <- df %>%
    select(c("marijuana-use", "n", "age"))
mar_18_21
```

```
## # A tibble: 4 x 3
##   `marijuana-use`     n age
##             <dbl> <int> <chr>
## 1            832.  2469 18
## 2            742.  2223 19
## 3            772.  2271 20
## 4            777.  2354 21
```

```r
mar_test_1819 <- prop.test(x = c(832.053, 742.482), n = c(2469, 2223))
mar_test_1920 <- prop.test(x = c(742.482, 772.14), n = c(2223, 2271))
mar_test_2021 <- prop.test(x = c(772.14, 776.820), n = c(2271, 2354))
# Printing the results

mar_test_1819
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(832.053, 742.482) out of c(2469, 2223)
## X-squared = 0.034721, df = 1, p-value = 0.8522
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.02448341  0.03048341
```

```
## sample estimates:
## prop 1 prop 2
##  0.337  0.334
```

```r
mar_test_1920
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(742.482, 772.14) out of c(2223, 2271)
## X-squared = 0.15514, df = 1, p-value = 0.6937
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.03408516  0.02208516
## sample estimates:
## prop 1 prop 2
##  0.334  0.340
```

```r
mar_test_2021
```

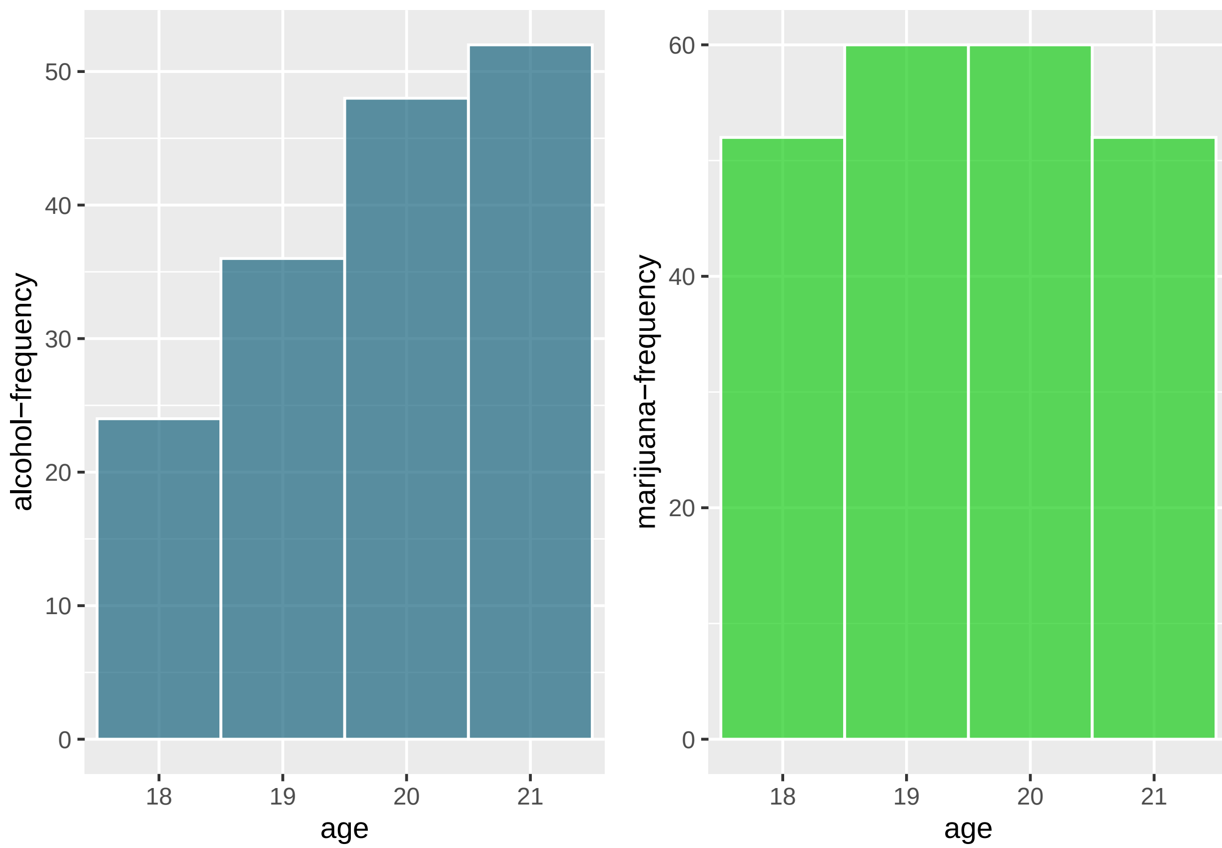
```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(772.14, 776.82) out of c(2271, 2354)
## X-squared = 0.475, df = 1, p-value = 0.4907
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.01764266  0.03764266
## sample estimates:
## prop 1 prop 2
##   0.34   0.33
```

```r
#From the output we can see the exact opposite for marijuana use and how the p-value
#is always greater than our alpha of 0.05. With this we can conclude that there is
#significant evidence to show that the difference in average usage from year-to-year
#is equal to 0 and does not increase or decrease. With 0 in the interval each year then the
#total change is still close to that of the start. Where the usage between 18 and 21 year
#olds is not that different compared to that of alcohol which is the opposite.
```

```r
alc_fre<-ggplot(df, aes(x=`age`, y=`alcohol-frequency`)) +
geom_bar(stat="identity", width=1, color="white",
         fill=rgb(0.1,0.4,0.5,0.7))

mar_fre<-ggplot(df, aes(x=`age`, y=`marijuana-frequency`)) +
geom_bar(stat="identity", width=1, color="white",
         fill=rgb(0.1,0.8,0.1,0.7))

grid.arrange(alc_fre, mar_fre, ncol=2)
```




```
#These graphs show data of how frequently (days of the year) alcohol and marijuana
#are used. We can see that alcohol graph is skewed left while the marijuana graph
#again is uniform. This is the same distribution that the usage followed, so it is
#interesting to see that an increase in new user usage also increases the frequency that
#the group of users also increases their use frequency.
```

CONCLUSIONS

After looking at graphs and numerical summaries of the data, my hypothesis of the data is incorrect in that the usage of marijuana and alcohol do not follow the same distribution from year-to-year. I was able to see that the average alcohol usage increases significantly each year, while marijuana tends to be the same each year. When examining the median number of times each drug is used in a year, the distribtions tend to look the same as well. I believe this might be from the popularity of drinking and how it may be easier to have new users versus that of marijuana who may find its target age around 18 and it doesn't add any new users. I think to lower usages in both categories, we can find ways to make sure to not get new users, during times where the data increases significantly.